

The width arm: additions to the golden report

Group G2, Tier 2

How to use this file

Each block below names the section of the golden report that it extends. It also says whether its text is inserted as new text or replaces existing text. Red markers stand for numbers and figures that wait on the width data. No number from the Tier 2 runs of 26 September 2026 is used. Those runs share one split across seeds, use the trace kernel and time grokking from memorisation, so none of their numbers holds under the definitions of the golden report.

Section, equation, table and figure numbers refer to the 21-page golden report of 4 October 2026.

Block 1. Section 1, Introduction

Where: insert as a new paragraph after the paragraph that ends “None of them measures the shape of the kernel.”

Width gives a test of these accounts that does not depend on the weight decay. In (1) the weight decay enters only the term $-2(k-1)\lambda\Theta_t$. That term is proportional to the kernel, so it changes the scale of the kernel and leaves its shape as it is. Only T_t can change the shape, and T_0 has expectation of order n^{-1} in the width n (Lewkowycz and Gur-Ari, 2021). If T_t stays of that order along training and shape change sets the grokking time, a wider network should grok later at a fixed weight decay. The grokking time of Kim (2026) scales as $(1-\beta)/(\eta\lambda)$ and holds the width at most through a logarithmic factor. We therefore add a width arm to the grid and ask which of the two the event times follow.

Block 2. Section 3.1, Task, network and training

Where: insert after the sentence that gives the readout factor $c_N = N^{-1/2}$ of the NTK parameterisation and $c_N = N^{-1}$ of the mean-field parameterisation.

Every width in this report uses the NTK factor $c_N = N^{-1/2}$. In the infinite-width limit the NTK stays constant and the network trains as a kernel method (Jacot, Gabriel, and Hongler, 2020), so with this factor a wider network is lazier at a fixed α . The mean-field factor keeps the dynamics non-degenerate as the width grows (Chizat, Oyallon, and Bach, 2020), and under it the width would not act as a setting of laziness. A width effect measured here belongs to the NTK parameterisation.

Where: replace the last sentence of the subsection, “The grid sets the product $\eta\lambda$ directly.”

The grid sets the product $\eta\lambda$ and the width N directly.

Block 3. Section 3.5, Event times

Where: insert after the sentence that ends “while (12) takes the pilot cell and the cells of the second stage together, $\eta\lambda = 0$ included, with $\eta\lambda$ itself.”

The width arm adds two fits of (10). The first takes the cells at the α of the pilot cell without weight decay, with the single covariate $\ln N$. The second takes the cells with $0 < \eta\lambda \leq$ **[TODO: largest level of the crossed arm]** at every width, with the covariates $\ln N$, $\ln \eta\lambda$ and their product. Both logarithms are centred at the middle of the grid before the product

is formed, so the first two coefficients are the slopes at that point. The width enters as $\ln N$. Indicators of its levels would fail at the wide widths. When every seed of a level is censored, the likelihood of an indicator for that level keeps rising as its coefficient grows, so the indicator has no finite maximum likelihood estimate. The model (12) takes $\ln N$ beside $\eta\lambda$. Its intervals resample seeds with all their runs, as above, so the runs of one seed at every width stay together. The relative risk model (13) is also fitted with $\ln N$ added to $Z(t)$. If the width acts on the event only through S_t , R_t and A_t , the coefficient of $\ln N$ in that fit should be close to zero (Kalbfleisch and Prentice, 2002).

Block 4. Section 3.6, Choice of levels

Where: insert as a new paragraph at the end of the subsection.

The levels of the width arm were not chosen by this design. The widths form a geometric grid with ratio 2, $N \in \{50, 100, 200, 400, 800, 1600\}$. The weight decays of the crossed arm, $\eta\lambda \in [\text{TODO: levels}]$, lie below the lower end $\eta\lambda = 5 \times 10^{-4}$ of the scan. That is the range in which the grokking time falls with the weight decay, and the fits of the second stage say nothing about it. Every cell of the width arm has a budget of 200 000 steps, twice the budget of the pilot cell and of the α arm. A width cell is compared with those arms at the shorter budget, by treating every event after 100 000 steps as censored there.

Block 5. New Section 4.2.3, Width

Where: insert as a new subsubsection after Section 4.2.2, Weight decay.

4.2.3 Width

This subsection changes the width of the pilot cell and holds every other setting fixed, with $\alpha = 1$ and $\eta\lambda = 0$. The widths are $N \in \{50, 100, 200, 400, 800, 1600\}$, each with **[TODO: number of seeds]** seeds and a budget of 200 000 steps. Table 1 gives the memorisation step and the grokking step at level 0.95 for each width, and Figure 1 the Kaplan-Meier estimate of the probability of not yet having grokked. **[TODO: State how the memorisation step changes with N , and how the grokking step changes with N . State at which widths no seed reaches each level within the budget.]** Table 2 gives the coefficient of $\ln N$ in (10) for each event and level.

Table 1: The sweep over width at the settings of the pilot cell. The memorisation level is training accuracy 0.99 and the grokking level is test accuracy 0.95. Medians, minima and maxima are over the seeds that reached the event.

N	Memorised	Grokked			
	Median	Median	Minimum	Maximum	Reached
50	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
100	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
200	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
400	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
800	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
1600	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

[TODO: Kaplan-Meier curves by width at level 0.95, drawn as in Figure 5, with Greenwood's exponential interval dashed.]

Figure 1: Kaplan-Meier estimates of the probability of not yet having grokked at level 0.95, by width at the settings of the pilot cell, with Greenwood's exponential interval dashed (Kalbfleisch and Prentice, 2002, Sec. 1.4.1).

Where: add these rows to Table 5, as the width arm.

Table 2: Rows of the width arm for Table 5: the coefficient of $\ln N$ in the log-normal accelerated failure time model (10), with its 95% interval, at the settings of the pilot cell.

Arm	Event	Level	Covariate	Coefficient	Lower	Upper
width	memorised	0.99	$\ln N$	[TODO:]	[TODO:]	[TODO:]
width	grokked	0.80	$\ln N$	[TODO:]	[TODO:]	[TODO:]
width	grokked	0.90	$\ln N$	[TODO:]	[TODO:]	[TODO:]
width	grokked	0.95	$\ln N$	[TODO:]	[TODO:]	[TODO:]

Block 6. New Section 4.2.4, Width and weight decay together

Where: insert as a new subsubsection after Block 5.

4.2.4 Width and weight decay together

Section 4.2.2 finds that above the break more weight decay delays grokking, and that its fits say nothing below $\eta\lambda = 5 \times 10^{-4}$, where the time falls. This subsection crosses that lower range with the widths of Section 4.2.3, at $\alpha = 1$, with $\eta\lambda \in$ [TODO: levels] and [TODO: number of seeds] seeds per cell. Figure 2 gives the log grokking time against $\ln \eta\lambda$ at each width, and Table 3 the coefficients of the crossed fit of (10). [TODO: State the signs of the coefficients of $\ln N$ and $\ln \eta\lambda$ and whether each interval excludes zero. State whether the product term is needed, and what its sign says about how the effect of weight decay changes with width.]

Table 4 gives the phase of every run at the last step, at the threshold 0.9 of Section 4.2.2. [TODO: State the largest $\eta\lambda$ at which each width still groks, and whether that edge moves as N grows.]

The model (12) with $\eta\lambda$ and $\ln N$ is fitted to every cell of the width arm, $\eta\lambda = 0$ included, and Figure 8 gains a panel for $\ln N$. [TODO: State whether the cumulative regression function of $\ln N$ rises, falls or stays flat along t .]

[TODO: Log grokking time at level 0.95 against $\ln \eta\lambda$, one line per width. Filled marks are observed events and open marks are runs censored at the budget, as in Figure 4.]

Figure 2: Log grokking time at level 0.95 against $\ln \eta\lambda$ for each width of the crossed arm. Filled marks are observed events and open marks are runs censored at the budget.

Table 3: Coefficients of the crossed fit of the log-normal accelerated failure time model (10) on the cells with $\eta\lambda > 0$, with their 95% intervals. The two logarithms are centred at the middle of the grid before their product is formed.

Event	Level	Covariate	Coefficient	Lower	Upper
memorised	0.99	$\ln N$	[TODO:]	[TODO:]	[TODO:]
memorised	0.99	$\ln \eta\lambda$	[TODO:]	[TODO:]	[TODO:]
memorised	0.99	$\ln N \cdot \ln \eta\lambda$	[TODO:]	[TODO:]	[TODO:]
grokked	0.95	$\ln N$	[TODO:]	[TODO:]	[TODO:]
grokked	0.95	$\ln \eta\lambda$	[TODO:]	[TODO:]	[TODO:]
grokked	0.95	$\ln N \cdot \ln \eta\lambda$	[TODO:]	[TODO:]	[TODO:]

[TODO: add the rows for the grokking levels 0.80 and 0.90.]

Table 4: Phase of the runs of the width arm at the last step, at the threshold 0.9, following Pracher et al. (2026). Each entry gives the number of seeds that grok, memorise, forget or do not fit.

[TODO: one row per width and one column per level of $\eta\lambda$, $\eta\lambda = 0$ included.]

Block 7. Sections 4.3 and 4.4, the kernel across width

Where: insert at the end of Section 4.3, The kernel at the grokking event.

Table 5 gives S_t , R_t and A_t at the grokking event at level 0.95 for each width without weight decay. S_t and R_t are taken on the kernel of the summed logits and A_t on the tangent kernel of all logits, as in (6). [TODO: State how each statistic at the event changes with N .]

Table 5: The statistics (6) at the grokking event at level 0.95, by width at the settings of the pilot cell. Medians over the seeds that grokked, with the minimum and maximum in brackets.

N	S_t	R_t	A_t
50	[TODO:]	[TODO:]	[TODO:]
100	[TODO:]	[TODO:]	[TODO:]
200	[TODO:]	[TODO:]	[TODO:]
400	[TODO:]	[TODO:]	[TODO:]
800	[TODO:]	[TODO:]	[TODO:]
1600	[TODO:]	[TODO:]	[TODO:]

Where: add to Section 4.4, What times the event.

Table 6 is fitted again on the runs of the width arm with $\ln N$ added to $Z(t)$ (Table 6).
[TODO: State whether the coefficient of $\ln N$ is close to zero once S_t , R_t and A_t are in the model, and which of the three carries the effect of width.]

Table 6: Coefficients of the relative risk model with time-dependent covariates (Kalbfleisch and Prentice, 2002, Eq. 6.14) on S_t , R_t , A_t and $\ln N$, with their 95% intervals and p -values, on the runs of the width arm.

Event	Level	Covariate	Coefficient	Lower	Upper	p
grokked	0.95	S_t	[TODO:]	[TODO:]	[TODO:]	[TODO:]
grokked	0.95	R_t	[TODO:]	[TODO:]	[TODO:]	[TODO:]
grokked	0.95	A_t	[TODO:]	[TODO:]	[TODO:]	[TODO:]
grokked	0.95	$\ln N$	[TODO:]	[TODO:]	[TODO:]	[TODO:]

[TODO: add the rows for memorisation and for the grokking levels 0.80 and 0.90.]

Block 8. Appendix B, Model checks

Where: insert after Figure 7.

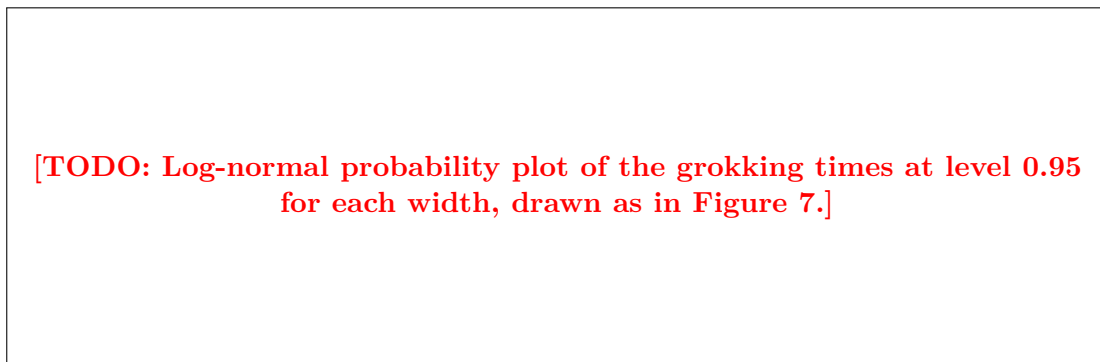


Figure 3: Log-normal probability plot of the grokking times at level 0.95 for each width of the width arm. Each observed time is placed at the normal quantile of its plotting position $(i - 0.5)/n$, and runs censored at the budget are left out (Nelson, 1990, Ch. 3, Sec. 4.2). Parallel straight lines support the log-normal model (10).

Decisions for the group

These points depart from the golden report or are not settled by it. The group should approve each one before the blocks above go into the master source.

1. The golden report has no width arm, and `CLAUDE.md` names a regression on width and weight decay as the head claim. The blocks above add width as a covariate of the report's own models. The group should confirm that the report takes this form.
2. The width arm has a budget of 200 000 steps, where the pilot cell and the α arm have 100 000. Block 4 says how the arms are compared.
3. The width arm has **[TODO: number]** seeds per cell, where the pilot cell and the α arm have 12.
4. The width levels and the weight decays of the crossed arm were not chosen by the design of Section 3.6.
5. The width enters the fits as $\ln N$ in place of indicators, for the reason given in Block 3.
6. The crossed fit uses only the cells up to **[TODO: largest level]**. The cells above that level are kept in the Kaplan-Meier estimates, in the phases and in the model (12). The choice selects cells, and every run in a chosen cell is kept, so it is consistent with Section 3.5. It must still be stated in the report.
7. Section 4.2.3 reports level 0.95. The golden report uses that level in every subsection except Section 4.2.2, which uses 0.90.